MPH — Job Allocation Scope, Objectives & Functionalities

Louta & Benguerir Mine Stockyards → MPH M16 Hub · M17 Sulphur — SP2M Program

Edition: Split report | **Date:** 17 September 2026 | **Source:** Maintained project scope

1. Context

The phosphate planning boundary is the complete **three-site value chain**, not M16 alone. Louta, Benguerir and MPH are peer controlled yards in **Domain A**; each has an SMS and receives one coordinated **seven-day plan containing hourly jobs**. An optional rolling mode refines the first four hours to 15-minute decisions and retains an hourly tail; timed asset changes and issued jobs keep exact-minute boundaries. Mining and washing upstream of the stockyards are interfaces, not directly controlled plant processes; the S0 simulator nevertheless represents selected Louta and Benguerir campaigns as auditable process-transfer stages so that ore placement, reclaim, recovery, product stacking and corridor/train withdrawals conserve mass. **Domain B** plans M17 sulphur independently on the same platform.



*** Relative proximity is user-confirmed (project team, 22-Jul-2026).** The 30/60 km reassignment and resulting 2/4-hour lags are provisional modelling values; exact conveyor route lengths, effective speeds and transfer dwell remain TBC.

Site / stage	Planning role / output	Principal open evidence
Louta beneficiation stock route · SMS	110LG ore storage → modelled L1/L2 transfer → 130/140LG product reclaim → **C1 → M16**; 102MG remains reference-only	Full GA/reach and route matrix; live stocks/assays; 102MG→110LG route/rate; approved case-to-product campaign map
Benguerir six-pile chain · SMS	3 mine hoppers → 2 mine-ore piles → 2 screened/raw piles → 3 washing campaigns → 2 B02 product piles → **C2 → M16 / train**	Physical tags and GA positions; approved Benguerir line rates/recoveries, grade mapping and train orders
MPH M16 · SMS	Receive, blend and reclaim → PAP or drying	Final GA, route confluences and metric separation limits
MPH M17 · SMS	Independently plan closed-store feed → melting	Demand/calendars; open reclaim stays manual

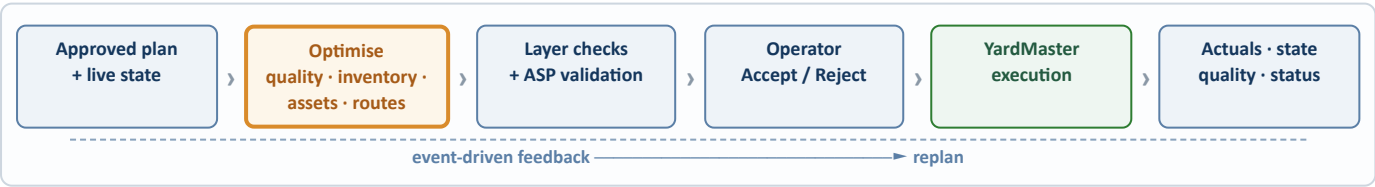
Phase 1 / expansion: M16 starts with PAP-1, 8 humid spaces and 3+2 machines. PAP-2 is already represented with the same demand template and the three explicit lifecycle states **Available / Disabled / Future**; its default is **Future**. Configuration supports N lines, 12 spaces and 4+3 machines. M17 closed storage expands from one to two stores and adds RC02. Mobile machines appear to be a separate package from the three solid-handling lots; final responsibility remains TBC.

System boundary: YardMaster/SMS maintains the 3D stock model and executes physical jobs. ASP supports manual authoring, rule validation and actuals tracking, but does not optimise demand, blends or multiple yards. This project is the **automated PlanAuthor above the three SMS**; ASP remains the vendor validation/execution bridge (§§14.2, 15.7–15.8).

2. Project Objective & Expected Impact

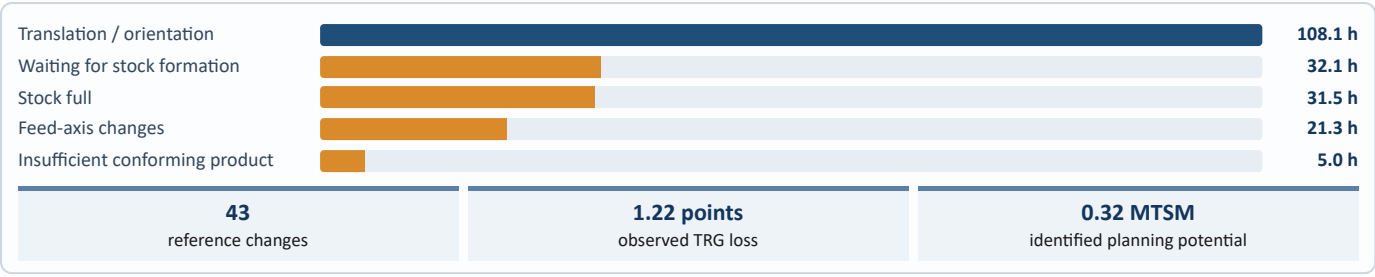
2.1 Objective

Turn approved dated demand and live yard state into an operator-supervised weekly rolling plan that coordinates quality, inventory, machines, conveyors, process lines, routes and corridor delays across Louta, Benguerir and M16; sulphur is solved separately. Each receiving unit may request several grade/quantity campaigns on the same day, with an intraday window and priority for each campaign.



2.2 Comparable OCP evidence — the planning loss already exists

The 2024 Khouribga Sidi-Chennane–Daoui diagnostic shows the same class of planning loss with only **2 stackers + 2 reclaimers**:



Comparable signal, not MPH forecast. Categories may overlap, Louta fleet data remain incomplete, and benefits must not be scaled linearly by machine count. Quantification requires de-overlapped replay, shadow mode and a supervised pilot.

2.3 Impact KPI scorecard

Operational targets remain **TBC** until an approved baseline, replay and pilot. Simulation may test **25% / 50% / 75%** recovery only on loss proven planning-addressable.

The Monitor makes these results explicit and shows the active objective weights beside the run. The current configurable objective includes unmet demand, shortage/safety exposure, movement, positioning, switching and quality deviation; the economic grade-cost term remains inactive until governed cost data exist. Weight values are scenario settings, not engineering facts, and the run stores them for reproducibility.

Area	KPI	Measure	Direction / fixed rule
Operations	Demand service	`Σ min(delivered,demand) / Σ demand`	↑ · TBC
	Feed interruption	demand > 0 and delivery = 0 [h]	↓ · TBC
	Corridor adherence	tonnes inside time/rate/quality window	↑ · TBC
Quality	On-spec mass	on-spec / delivered tonnes	↑ · TBC
	Off-spec exposure	off-spec tonnes + weighted deviation	↓ · TBC
	Give-away	weighted excess above target	↓ · TBC
Inventory/assets	Reconciliation drift	planning twin − SMS inventory	↓ · TBC
	Inventory bounds	time inside safety/capacity limits	↑ · overcapacity **0**
	Planning-addressable loss	de-overlapped hours + moves/starts per kt	↓ · 25/50/75% sensitivity
Planning	First-pass ASP	AUTO items with zero `ERROR` s	↑ · TBC
	Automation coverage	accepted AUTO / eligible	↑ · TBC
	Routine effort	operator authoring minutes / shift	↓ · TBC
Resilience	Replan latency	event → valid replacement	↓ · TBC
	Service recovery	disruption → restored feed	↓ · TBC
Governance	Schedule integrity	invalid/conflicting jobs	**0**
	Human-plan preservation	human items changed	**0 / 100%**
	Deterministic replay	same inputs/config/seed	**exact**
Integration	SMS ownership integrity	unmapped/wrong-SMS jobs	**0**

Claim boundary: baseline → offline replay → shadow mode → supervised pilot → sustained/economic validation. No throughput, financial, energy, staffing, reliability or safety benefit is claimed before pilot evidence. Formal acceptance measures are in §18.

3. Where this Layer Sits

Tier	Owner	Responsibility
Weekly / monthly planning	Operations/planning	Demand, campaigns, strategic stocks and expected supply; the application edits the approved seven-day demand envelope
Job Allocation Layer	**This project · L3**	Quality/quantity, blend, pile, machine, route and sequence
ASP + YardMaster/SMS	Vendor execution system	Validate, execute, maintain 3D state and return actuals
PLC / DCS / ACS	Control systems	Commands, interlocks, anti-collision and safety

Replanning is event-driven on quality drift, stock change, late supply, demand change, machine fault or execution deviation—not a fire-and-forget schedule.

4. Scope Boundary

In scope	Out of scope / retained authority
Coordinate a seven-day plan of hourly or mixed-resolution jobs across the 3 phosphate SMS; plan M17 independently	Generate the upstream monthly forecast or approve business demand
Select quality, quantity, blend, pile, machine, route and sequence	Mining, washing, PAP, drying and melting process operation
Direct target space when segregation requires; otherwise permit SMS auto-route	Physical actuation, deposition, 3D model and anti-collision
Track actuals, replan, explain and support what-if	Build/own SMS, SQMS, analysers, LIMS, CMS or equipment

Capability	Allocation layer	ASP	YardMaster / PLC / ACS
Multi-yard demand/blend optimisation	**Owns**	—	—
Physical route/shared-resource feasibility	**Owns** (§14.2)	Validates configured tuple	Executes route
Plan-item validation	Pre-checks	**Second gate**	—
Accept/reject/override	Preserves humans	Supports workflow	Executes accepted job
Safety/interlocks	Never replaces	—	**Retained authority**

An ASP "Route" is only `(Device, JobType, Location)`; it is not the conveyor network. RabbitMQ access/licensing and site master-data mappings remain TBC.

5. Functional Scope

Inputs	Decisions	Outputs
Dated demand campaigns: unit, grade, physical quantity, intraday window and priority	Source quality and quantity	Prioritised jobs per SMS/machine
SMS 1-m inventory, quality and age	Blend mode, component masses/shares and destination	Stack: source, grade, method, target pile + repère range, rate, tonnes
Actuals and machine calendars	Machine, pile, route and timing	Reclaim: pile + repère range, route, blend, rate, tonnes
Analyser + lab results	Reallocation on deviation	Profiles, alerts, rationale and KPIs

5.1 Repère position model

Each stockpile uses the operating scale **R0–R1000**, representing **1,000 equal longitudinal intervals**. A job may therefore state "store this grade on `LT-P1` from **R400 to R600**." Repères are operator-facing; physical metres remain canonical for SMS/ASP exchange, inventory segments, machine travel and safety distances:

$x_m = x_0_m + (R / 1000) \times (x_1_m - x_0_m)$ · $R = 1000 \times (x_m - x_0_m) / (x_1_m - x_0_m)$

The Innomotics FRS confirms numeric Start/End positions (`StartX` / `StopX`), per-metre pile inventory and position-range validation. The term **repère** and the R0–R1000 scale are a **project-team operating convention confirmed 24-Jul-2026**, not a claim that every pile is 1,000 m long. Before live use, each pile still needs its R0 endpoint and increasing direction, surveyed physical origin/usable length, and the ICD's StartX/StopX units and scale.

Constraint family	Planning rule
Corridors	C1/C2 capacity, availability, lag and in-transit inventory
Quality	Destination intervals, anti-contamination and stage-correct blending; dry-solids weighting for chemistry and physical-mass weighting for moisture
Inventory	Initial mass, safety stock, capacity and conservation
Machines	Reachability, exclusivity, rate, travel/setup, maintenance and rails
Same-pile work	M16 concurrency requires spatial/work-zone/route separation; limits TBC; mine applicability TBC
Sulphur	Open reclaim manual; closed RC01 and ≤8 start-stops/h planned
Dry stream	Dryer intake bounds humid reclaim; drying OOS

Detailed variables, constraints and equations are defined once in §§15.1–15.4.

6. Architecture & Interfaces

L4 · Business	ERP / production plan · Historian / reporting	managed OT/IT DMZ
L3 · Planning	**JOB ALLOCATION LAYER ↔ Operator HMI · LIMS · CMS**	
L2 · Control	ASP / SMS ×3 · SQMS · PLC / DCS / ACS	jobs ▼ · actuals/state ▲
L1 · Field	Stackers · reclaimers · trippers · gates · analysers · belt scales	

Counterpart	Dir.	Minimum exchange	Mechanism / cadence
SMS ×3	↔	Plan items; ASP issues; `JobUpdated`; 1-m inventory/quality/age; machine state	Common YardMaster/RabbitMQ profile; per-site IDs TBC · event/continuous
SQMS / analysers	→	BPL, Cd, MgO, SiO_2, CaO/CO_2, moisture	OPC UA · continuous
LIMS	→	Detailed assay and PSD	API · per sample
ERP / planning	→	Approved demand, supply and campaigns	MES API/file · change
CMS	→	Availability and maintenance	API/OPC UA · event
Historian / HMI	↔	Trends, KPIs, plan, rationale and overrides	Historian + web/HMI · continuous

Deployment: dedicated L3 application on controlled OT; business exchanges cross a managed DMZ. Missing/stale SMS data are flagged with last-good-state plus operator fallback. **Configured:** sites, assets, reachability, route graph, specs, compatibility, patterns, weights, calendars, endpoints and IDs.

> All equipment, conveyor and stockpile capacities and rates are represented in **physical/as-handled tonnes** at their material stage. TSM is a reported merchant-dry equivalent, not the capacity basis: raw physical mass converts through raw moisture and beneficiation yield; wet washed-product physical mass retains product moisture; after drying, 1 physical dry tonne = 1 TSM. The unified Grade configuration page maintains chemistry, yield/moisture, material profiles and receiving limits, and supports persistent add/delete with referential checks.

7. Representative Use Cases

Use case	Trigger	Decisions	Jobs	Replan
PAP on-grade humid feed	Dated campaign + pile quality	Build an eligible humid stockpile with fixed-matrix or optimized shares, then choose reclaim rate, machine and route	Component stacking followed by ready-boundary-safe reclaim	Cd/BPL, asset or demand drift
Drying / SPH	Spec + dryer limit	Build pile, humid blend, route	Stack/reclaim/gate	Quality, dryer or rail change
Sulphur continuity	Melting demand + closed stock	Replenishment and RC01 rate	Tripper/stack/closed reclaim	Delivery/payloader/RC01 deviation

8. Assumptions, Dependencies & Constraints

ID	Current ruling	Residual action / hedge
H1	Telegram access feasible in concept	Obtain ICD, credentials, licences, namespaces and keys; otherwise standard ASP client path
H2	**Resolved at solution level:** all sites use YardMaster	Configure three endpoints, IDs, routes, calendars and permissions
H3	Layer proposes; humans supervise; ASP arbitrates	`AUTO-` ownership only; human items fixed; audit retained
Data	Inputs assumed timely enough	Freshness limits, last-good fallback and alarms
Demand	Approved input-side demand authoritative	Validate source, cadence and governance

9. Deliverables & Phasing

Phase	Outcome	Exit evidence
0 · Scope/data	Boundaries, evidence and interfaces	Approved scope + gap owners
1 · Functional design	Twin, optimisation, constraints and ICDs	Design review + acceptance set
2 · Twin/shadow	Replay, simulation and supervised shadow	Baseline, validation and operator sign-off
3 · Deployment	Three-SMS closed loop and KPIs	Supervised pilot then acceptance

10. Key Risks

Risk	Control / proof
Stale/inaccurate quality or inventory	Freshness, reconciliation, fallback and actuals replay
Missing SMS ICD/access/master data	Vendor workshop, emulator and per-site mapping tests
Unreliable demand/route model	Input governance, route audit and zero-invalid-route acceptance
Asset/layout mismatch	GA/equipment list, reachability review and shadow validation
Operator distrust/conflicting edits	Rationale, Accept/Reject, human-item preservation and audit
Overstated benefits	De-overlap, approved baseline and no pre-pilot extrapolation

11. Success Criteria

The canonical outcome scorecard is §2.3; technical acceptance is §18.

Historical baseline › **Offline replay** › **Shadow mode** › **Supervised pilot**

12. Non-Functional Requirements

Requirement	Design intent	Status
Latency	Event-to-valid-replan fast enough to avoid avoidable gaps	TBC
Availability	Redundancy; last-good-state + operator fallback	TBC
Data integrity	Freshness, completeness, reconciliation and traceability	TBC
Scalability	Expansion/new areas through configuration	Required
Security	OT/IT segregation, RBAC, least privilege and audit	Required
Usability	Rationale, override, what-if and evidence status	Required
Maintainability	Versioned configuration, deterministic tests and simulation	Required

13. Stockyard Layout, Assets & Evidence Gaps

This is the canonical layout/model-coverage summary. Detailed provenance and conflicts remain in §16; positions stay schematic until GA/plot-plan data are received.

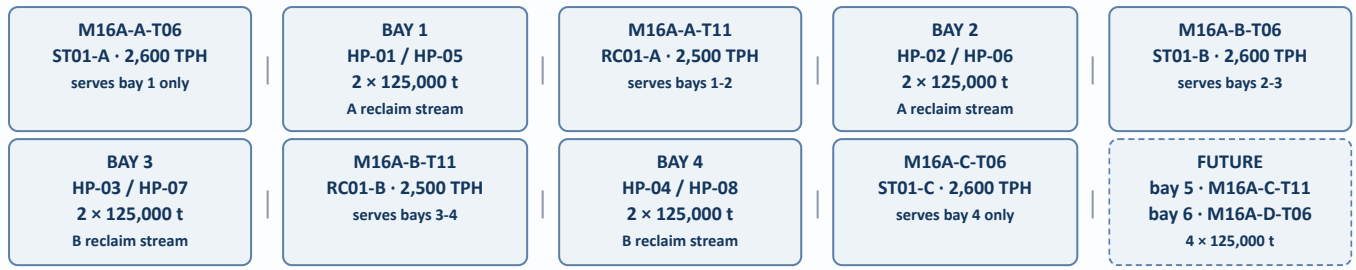
13.1 Canonical site / asset register

Site / stage	Material / storage	Stack assets	Reclaim assets	Route	Model status
Louta 102MG mine storage	MINE STOCKPILE-01/-02; **ASSUMED display baseline:** 2 × 100,000 t capacity, 50,000 t t0 each (50%), two-profile 50/50 bands; actual capacity/live state TBC	102MG-ST-01/-02 visible	route/rate/reach TBC	intended transfer toward 110LG; never direct to C1	**Reference only:** `MODELLING_GAP[MISSING_ROUTE_RATE_REACH]`; excluded from solver inventory, jobs and KPIs

Site / stage	Material / storage	Stack assets	Reclaim assets	Route	Model status
	— the 494,274 t ore capacity is *not* theirs				
Louta 110/120LG homogenized ore	2 stockpiles · 494,274 t total · 2 × 247,137 t **DERIVED** equal split · 50% start = 123,568.5 t each (247,137 t total, ASSUMED)	1 shared 110LG-ST-01 · 2,100 TPH wet · chevron via boom conveyor 110LG-BC-01	2 bridge reclaimers 120LG-RC-01/-02 · 3,000 TPH wet each	RB1–RB3 → common RB4/RB5 → washing L1 or L2	**Solver-backed S0 route:** both ore lanes and all three machines are planned; displayed multi-profile bands remain non-spatial metadata
Louta washing / beneficiation	Scrubber residence **2.5–3 min CONFIRMED**; overall washing-line residence **0.5 h (30 min) ASSUMED**	2 parallel production lines **L1 + L2**, 3 Mt tsm/year each; solver process-transfer stages, not mobile machines		recovered WIP → 130LG product stacking	**Solver-backed process transfer:** documented case recovery is applied; no same-bucket passthrough and cumulative matured-mass availability enforce the assumed 30-min dependency
Louta 130/140LG product	383,024 t total · 2 × 191,512 t **planning/solver lines** (DERIVED equal split); starting mass 95,756 t each (50%) — physical pile count not stated in the source	1 shared 130LG-ST-01 (fed by 130LG-CV-11/SL11); **1,000 TPH S0 envelope ASSUMED**	140LG-RC-01/-02 · 3,000 TPH · conservative one-per-planning-line pairing	RL1/RL2 → flip gates 140LG-FG-01/-02 → RL3/RL4 → C1	**Solver-backed S0 route:** stack, product inventory, reclaim and exact 8,000 t C1 contribution are planned
Benguerir mine-ore storage	`BG-MINE-1/2` , 2 × 200,000 t **ASSUMED S0 capacity**, 100,000 t start each; deliberately larger than Louta's 2 × 100,000 t 102MG display baseline	`BG-MINE-ST-1/-2` , one per pile · 2,100/2,500 TPH borrowed envelope	shared `BG-MINE-BWR` · 2,100/2,500 TPH borrowed envelope	`TM1` / `TSM2` / `TSM3` handoffs → mine piles → screened transfer	**Solver-backed:** inventory and stack/reclaim jobs planned; names, reaches and rates are assumptions pending tags/GA
Benguerir screened/raw storage	`BG-RAW-1/2` , 2 × 200,000 t (400,000 t Table-20 total), 100,000 t start each · <15 mm	`BG-RAW-ST-1/-2` , one per pile · 2,100/2,500 TPH	shared `BG-RAW-BWR` · 2,100/2,500 TPH	three washing campaigns; raw-train interface TBC	**Solver-backed:** both piles and all three machines participate in conserved process transfers
Benguerir washing / beneficiation	Scrubber residence **3 min CONFIRMED**; overall line residence **0.5 h (30 min) ASSUMED and modelled**	`BEN-WASH-L1/L2/L3` , all three OPERATING/configurable in S0; process stages, not mobile machines		recovered WIP → shared ST3 → B02	**Solver-backed:** three demand-driven feed/recovery campaigns; rates, recoveries and grade mapping are borrowed/ASSUMED
Benguerir B02 washed	`BG-P1/P2` , 2 × 191,512 t · settled product	current twin-boom `ST3` · 2,100/2,520 TPH	current `RC2/RC3` · 2,600/3,120 TPH; S0 pairs RC2→P1 and RC3→P2	C2 → M16; product train	**Solver-backed:** ST3 stack, pile inventory and RC2/RC3 reclaim jobs are scheduled from demand
M16 humid	8 × 125,000 t in **4 side-by-side two-hangar bays** (+ 4 × 125,000 t future); hangar names/left-right order INFERRED	ST01-A/B/C · ≈2,600 · one per line `M16A-A/B/C-T06`	RC01-A/B · ≈2,500 · `M16A-A/B-T11`	2 separate gate banks `M16A-A/B-H01…H04` → PAP or drying (confluence TBC)	Modelled; each shared line visibly spans its horizontal pile pair; machine reach remains bay-adjacent
M16 dry	2 × 50,000 t	trippers	gravity gates	rail→SPH	Interface abstraction
M17 open/closed	2×100,000 / 1×100,000 t	ST01 / T04T	manual / RC01 ≈1,100	melting	Manual open; closed modelled

Layout rulings: M16 is **four two-hangar bays with the machine lines interleaved between them** — `M16A-A-T06` (ST01-A) · bay 1 · `M16A-A-T11` (RC01-A) · bay 2 · `M16A-B-T06` (ST01-B) · bay 3 · `M16A-B-T11` (RC01-B) · bay 4 · `M16A-C-T06` (ST01-C) — per `Q80130-MZI-M16-ME-PFD-0001[3]` Rev 3. In the mimic, each bay is explicitly drawn as **one side-by-side horizontal pair**, with every serving machine line spanning both hangars (project-team display ruling, 26-Jul-2026); this makes shared access legible without changing model reachability. There is **no shared T06/T11 corridor** and no detached machine corridor. A machine translating between adjacent bays serves **only** those bays (project-team rule, 25-Jul-2026): ST01-A→bay 1 · RC01-A→bays 1-2 · ST01-B→bays 2-3 · RC01-B→bays 3-4 · ST01-C→bay 4; an unreachable job is rejected by validation and by the ASP work-area rule. Same-pile stack/reclaim is permitted subject to separation and work-zone rules, with numerical values TBC. Hangar names and left/right order remain INFERRED; metric coordinates remain schematic (the PFD is a flow sheet, not to scale); mine-site coordinates and concurrency applicability remain TBC.

13.2 M16 — side-by-side hangar pairs with machine lines between bays

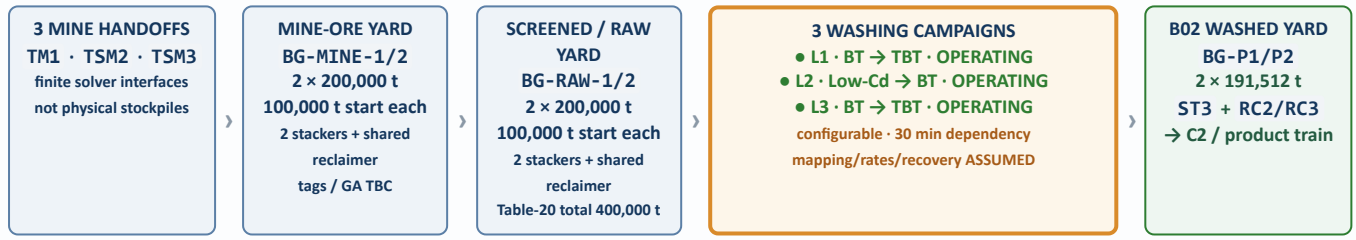


Reclaim take-off — two separate streams: each reclaimer feeds its own `T11` → `T11S` magnetic separator → `T12` (belt scale) → hopper `M16A-A-R01` with air blasters → **its own clamshell + rod gate bank** (`M16A-A-H01…H04` / `M16A-B-H01…H04`) → `T13` outbound belt at 1,500 TPH. **That chain is confirmed equipment.** What is *not* confirmed is the destination: neither the PFD nor PRO-0001 says which gate or downstream diverter sends a stream to PAP rather than to drying, and the confluence point is not resolvable from the available sheet. The model therefore separates the two — a **confirmed** gate-bank link per stream, then a **derived/TBC** route selection toward PAP (six parallel 1,500 TPH conveyors, 3 running + 3 future) or drying (two parallel 1,299 TPH conveyors, both out of scope). One destination per bank per bucket is derived from the single outbound belt; the documented 10-minute changeover is recorded but **not modelled**.

Reception envelope — one current path: PRO-0001 §2.1.1 places auto sampler `M16VP01` on `M16A-A-T01` and `M16VP02` on "the adjacent **future** conveyor", and the PFD marks that line FUTURE. C1 and C2 therefore **share one current 2,600 TPH reception path**: aggregate reception is capped at 2,600 TPH, never 2 × 2,600 TPH, and the initial in-transit profile shows the corridors running alternately rather than both at full rate in the same hour.

> **What is confirmed and what is not:** the eight 125,000 t hangars, the four-bay/machine-line sequence, the machine identities, their lines and their rates are **confirmed** by `Q80130-MZI-M16-ME-PFD-0001[3]` Rev 3. The **hangar names inside a bay are inferred** — the PFD does not name them, so `HP-01…08` remains a working convention pending the plot plan and the official SMS pile-ID register. Metric positions stay schematic: the PFD is a flow sheet, not to scale. Machine-to-bay assignment applies the project-team rule that a machine translating between adjacent bays serves those bays only; per-line travel limits still require a surveyed GA.

13.3 Benguerir — six physical piles with three solver-backed washing campaigns



The six physical stockpiles are `BG-MINE-1/2`, `BG-RAW-1/2` and `BG-P1/P2`. The three hopper rows and line WIP buffers are finite solver interfaces, **not additional on-ground stockpiles**. The mine-ore, screened and B02 stages are connected at process level but use their own pile and machine resources. B01 `04BG-*` and B02 `RB/SL/RL-*` physical tags are not merged without PFD reconciliation.

Evidence / issue	Status and consequence
Table 20 screened-yard topology	**User-confirmed** 20-Jul-2026; retained as `BG-RAW-ST-1/-2` + `BG-RAW-BWR` and promoted into the planning model
Six-pile capacity / starting mass	**Physical S0 spaces:** `BG-MINE-1/2` = 200,000 t capacity and 100,000 t start each (**ASSUMED**); `BG-RAW-1/2` = 200,000 t capacity and 100,000 t start each (400,000 t Table-20 total); `BG-P1/P2` = 191,512 t each. The first mine-ore pair is intentionally larger than Louta's assumed 100,000 t 102MG piles. Timestamped SMS telemetry is not loaded.
Three mine hoppers	`TM1`, `TSM2` and `TSM3` are represented by three finite 10,000 t planning handoffs. This is a campaign buffer, **not a claimed physical hopper capacity or live reading**; exact tags/capacities/routing remain TBC.
Raw-grade and line mapping	Benguerir and Louta share the mine-neutral grade IDs `RAW-BT`, `RAW-BT-HIGH-CD` and `RAW-BT-LOW-CD`; the mine is not encoded in the grade name. S0 uses L1/L3 BT→TBT and L2 Low-Cd→BT. Chemistry, recoveries and mapping are **ASSUMED simulation inputs borrowed from Louta design cases**, not live Benguerir assays or approved recipes.
Mine-ore machine identities and reach	`BG-MINE-ST-1/-2` and shared `BG-MINE-BWR` are user-directed S0 planning assets. Their 2,100/2,500 t/h envelopes are borrowed; physical tags, surveyed reaches and metric GA coordinates are **TBC**.
Table 20 doc no./rev, remaining tags and coordinates	**TBC**; none invented
2,100/2,500 vs 2,600/3,120 TPH	**Conflicting**; raw→C2 at 2,600 assumes resolution
ST3/RC2/RC3 and **	**Current configuration and solver-backed**; notation only. S0 plans recovered-product stacking through ST3, then pairs RC2→BG-P1 and RC3→BG-P2 for demand withdrawals.
Washing-line planning baseline	`BEN-WASH-L1/L2/L3` are all **OPERATING and configurable** in S0. Commissioning/as-built lifecycle evidence remains TBC.
Washing-step timing	**Scrubber residence = 3 min CONFIRMED** by `Q80130-BEN-B02-PR-BOD-0001` Rev 0, §7.1 p. 19 and §7.3.1 p. 25. The separately instructed **ASSUMED overall line residence = 0.5 h (30 min)** is enforced as the S0 wash-to-ST3 dependency; the 3 min component is not added again.

Evidence / issue	Status and consequence
“3 qualities on 2 piles”	Zoning/campaign/compatibility rule **TBC**
Train routes/station	Interfaces documented; ownership/sharing **TBC**
B01 rates	541+89=630 vs 628; 1,208 vs 1,055 · unresolved

Simulator coverage: all six physical Benguerir piles, all nine mobile stockyard machines, all three mine-hopper handoffs and all three washing campaigns are connected to the planning model. A campaign plans hopper-to-mine stacking, mine reclaim, screened-pile stacking, washing reclaim/recovery and recovered-product stacking through `ST3`; configured train or C2 demand then creates `RC2` / `RC3` reclaim work. The former continuous `BG-WASH-IN` shortcut is retired and inactive. Hopper and WIP spaces are solver interfaces rather than physical stockpiles. OPERATING and availability are scenario configuration, not live telemetry.

Mimic inspection contract: Initial state shows scenario t0; **Final state** shows terminal plan inventory after all horizon buckets. Stockpile fill length is proportional to physical occupancy rather than stretching grade bands to the pile outline. Hover exposes grade composition, physical mass, capacity/fill, material state and stage-correct TSM equivalent when a conversion profile exists. Hopper and WIP interfaces do not appear as extra physical stockpiles.

13.4 Material and grade states



Material state and grade are separate model dimensions. A washed grade vector must never be reused for raw/screened phosphate.

Stage	Documented physical profile	Chemical / grade status	Model profile
Benguerir mine/screened	<15 mm · 10–14% moisture · 1.2–1.4 t/m³ · 30–35°	Shared mine-neutral raw IDs; S0 selects BT on lanes 1/3 and Low-Cd on lane 2. No live/pile-specific Benguerir assay or cut-offs	`BEN-RAW-SCREENED` · solver-backed BG-MINE-1/2 and BG-RAW-1/2
Benguerir filtered	23% moisture · 1.2–1.4 t/m³ · 35°	Assays/cut-offs TBC; S0 recovered mass derives from borrowed case recoveries	`BEN-WASHED-FILTERED` · finite line WIP interface
Benguerir settled	15–20% moisture · 1.2–1.4 t/m³ · 35°	TBT/BT/MT names; S0 produces TBT/BT, while site limits and approved pile rules remain TBC	`BEN-SETTLED-PRODUCT` · solver-backed BG-P1/P2
Louta mine/product	Raw: 15% moisture · SG 2.9 · F100 10 mm; combined product: moisture <22% · SG 3	Raw BT profiles feed assumed 102MG/110LG display compositions; LT-P1/P2 use assumed washed TBT/BT/MT distributions; operating mapping/cut-offs TBC	Separate raw reference and product-planning profiles
M16 humid	PGNAA + moisture + SMS/SQMS	Grade/SQP IDs/limits TBC	`M16-WASHED-HUMID`

Demo grade	BPL	Cd ppm	MgO %	SiO_2 %	Moisture %	Status
TBT	74	22	0.8	2.0	12	washed simulation placeholder
BT	68	30	1.1	2.8	12	washed simulation placeholder
MT	63	35	1.4	3.5	12	washed simulation placeholder

Louta product-yard demonstration distribution

Product pile · starting state	Assumed grade distribution	Mass-weighted composite used by planning
LT-P1 · 95,756 / 191,512 t (50%)	60% TBT + 40% BT	BPL 71.6% · Cd 25.2 ppm · MgO 0.92% · SiO_2 2.32% · moisture 12%
LT-P2 · 95,756 / 191,512 t (50%)	60% BT + 40% MT	BPL 66.0% · Cd 32.0 ppm · MgO 1.22% · SiO_2 3.08% · moisture 12%
Yard aggregate · 191,512 / 383,024 t (50%)	30% TBT + 50% BT + 20% MT	BPL 68.8% · Cd 28.6 ppm · MgO 1.07% · SiO_2 2.70% · moisture 12%

TBT/BT/MT are **ASSUMED washed-product demonstration vectors**; their operating Louta grade mapping and cut-offs remain TBC. Band widths show mass shares only—not surveyed spatial order, stacking chronology or reclaim sequence. For these Louta demonstration piles, planning uses each pile's aggregate composite quality and does not select a component profile or zone independently. Profile-selective reclaim requires surveyed boundaries and the active reclaim face; the implemented M16 pile-construction blend is the separate, explicitly bounded case in §13.7. The shared `RAW-BT-*` grades are a **separate material state**. Louta's display compositions remain assumed metadata. In the Benguerir solver chain they are campaign feed identifiers: BT feeds L1/L3 and Low-Cd feeds L2 before the assumed wash transformations produce TBT/BT. Raw-grade vectors are never applied directly to washed-product inventory.

Shared raw BT grades — current chemistry source: Louta L02 design basis

Profile / specification	BPL %	MgO %	CO_2 %	SiO_2 %	Cd ppm	Moisture %	SG	PSD
BT raw — component of both demonstration piles	56.06	1.49	8.21	13.71	21	15	2.9	F100 10 mm
High-Cd BT raw — P1 component	55.11	1.50	8.41	16.28	25	15	2.9	F100 10 mm
Low-Cd BT raw — P2 component	58.28	1.13	8.75	10.22	13	15	2.9	F100 10 mm
Combined product specification — reference only	68.29	<0.6	6.81	<4	17	<22	3	—

Louta display-only composite chemistry and Benguerir S0 feed assignments

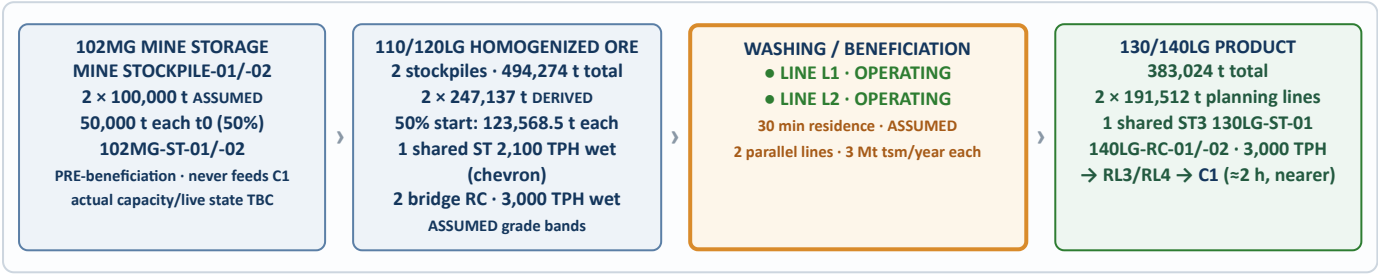
Raw pile · assumed mass composition	BPL %	MgO %	CO_2 %	SiO_2 %	Cd ppm	Moisture
102MG-SP-01 · 50% BT + 50% High-Cd	55.585	1.495	8.31	14.995	23	Louta 15%
102MG-SP-02 · 50% BT + 50% Low-Cd	57.17	1.31	8.48	11.965	17	Louta 15%
110LG-OL1 · 50% BT + 50% High-Cd	55.585	1.495	8.31	14.995	23	Louta 15%
110LG-OL2 · 50% BT + 50% Low-Cd	57.17	1.31	8.48	11.965	17	Louta 15%

BG-MINE-1 / BG-RAW-1 · BT S0 campaign	56.06	1.49	8.21	13.71	21	Benguerir 10–14% physical metadata
BG-MINE-2 / BG-RAW-2 · Low-Cd S0 campaign	58.28	1.13	8.75	10.22	13	Benguerir 10–14% physical metadata

Source: `Q80130-LOU-L02-PR-BOD-0001` Rev B, 12-Apr-2023, §6.1 pp. 14–15 and p. 18, plus project-team multi-profile and shared-grade clarifications through 28-Jul-2026. The grade IDs are shared across the mines, but the numeric chemistry used for the Benguerir S0 campaign is borrowed from the Louta design basis and remains **ASSUMED**, not a Benguerir assay. Louta's R-band shares remain schematic display metadata; Benguerir's solver-backed pile state uses its configured campaign grade and live/terminal tonnage. The combined-product row retains the document's own stage wording and does not replace the TBT/BT/MT demonstration vectors.

Needed: pile survey or SQMS segment data to replace the assumed shares, band widths and mapping; live/pile-specific Benguerir raw assays and raw/washed min-typ-max assays by site/grade; Grade/SQP IDs and cut-offs, sampling basis, PSD/moisture, pile/zone campaign/compatibility rules and SMS mappings. Live SMS/SQMS values override the demonstration composition and current design-basis vectors.

13.5 Louta — four stages from mine storage to the C1 corridor



Applied planning geometry, not documented pairing: the process description confirms two ore stockpiles and two ore reclaimers but never states which reclaimer serves which pile — both reclaim routes converge on the common RB4/RB5 transfer that can feed either washing line. The mimic therefore draws one reclaimer per pile as the applied planning arrangement. The same caution applies downstream: the **physical** product-pile count is not stated, so LT-P1/LT-P2 are planning/solver lines with a derived equal split.

> **Mimic inspection contract (Louta): Initial state** shows 102MG mine storage at 50,000 t per pile, 110LG ore at 123,568.5 t per pile and product at 95,756 t per line — all 50%. **Final state** shows terminal plan inventory for the solver-backed 110LG and product lines. Only 102MG stays at its static assumed reference baseline in both views.

Authoritative equipment source: **Louta Beneficiation Plant Process description, Rev B (IFA), 25 April 2023, §3.1 pp. 5–8. Document-control inconsistency recorded:** the cover states `Q80130-LOU-L02-PR-REP-00002` while every body header states `Q80130-LOU-L02-PR-REP-00004`; both are retained and neither is silently preferred. Storage totals come from the **Louta BoD Rev B**, which gives no individual pile capacities. BoD p.20 lists one ore and one product reclaimer, while the later tagged process description identifies `120LG-RC-01/-02` and `140LG-RC-01/-02`; S0 uses the tagged two-plus-two configuration and preserves both count conflicts. Neither PDF contains `102MG` or the label `C1` (the process description calls the outbound route `BL`).

> **S0 stocked-route policy — ASSUMED project policy, not a design-document operating rule.** Exact configured targets are C1 dispatch **8,000 t**; recovered output **4,000 t from L1 + 4,000 t from L2**; terminal `110LG-OL1/-OL2` **123,568.5 t each**; terminal `LT-P1/-P2` **95,756 t each**; and line WIP **0 t**. L1 uses documented BT feed/recovery **584 t/h / 68.22%**; L2 uses Low-Cd **588 t/h / 67.71%**; High-Cd **579 t/h / 68.83%** remains documented but unselected. BT→L1→TBT and Low-Cd→L2→BT are **ASSUMED mappings**. `MACHINE_COVERAGE_REWARD=0` and the positive commanded-tonne cost remove coverage rewards and token jobs.

The S0 source interface uses two finite **10,000 t ASSUMED mine-handoff inventories**; they are not 102MG. Common RB4/RB5 is conservatively capped at 588 t/h with one ore reclaimer at a time; the unresolved RL3/RL4/C1 confluence permits one product reclaimer at a time. The 130LG stacker envelope is **1,000 t/h ASSUMED** because no source rate is supplied. Direct ore and product storage bypasses are documented, so idle equipment may be legitimate. Solver output reports scheduled work or structured dispositions: unavailable, no inventory, no target/free capacity, missing route, shared-route peer selected, or target met without that machine. Neither reclaimer pair is identified as duty/standby in the PDFs.

Stage	Known	Not established	Required next
102MG mine storage	MINE STOCKPILE-01/-02 and visible 102MG tags; pre-beneficiation and never direct to C1. S3-A25 displays an assumed 100,000 t capacity / 50,000 t t0 per pile	Actual capacity/live inventory, grade order, duties, reach, route/rate and metric positions. No supplied Louta PDF contains 102MG or establishes its connection to 110LG	Full drawing, GA, route/rate/reach matrix and timestamped SMS/SQMS snapshot; remains `MODELLING_GAP[MISSING_ROUTE_RATE_REACH]`
110/120LG homogenized ore storage	Two solver-backed lanes; shared 110LG-ST-01 at 2,100 TPH wet; 120LG-RC-01/-02 at 3,000 TPH wet; common RB4/RB5; total 494,274 t	Equal pile split DERIVED; start/terminal state, grade bands and conservative pairing ASSUMED. BoD says one ore reclaimer; later tagged description says two, which S0 uses	Live stocks/assays, pile schedule, reach/GA, common-route capacity/concurrency and approved campaigns
Washing L1 / L2	Two 3 Mt TSM/year lines. BT 584 t/h / 68.22%; High-Cd 579 / 68.83%; Low-Cd 588 / 67.71% are documented. Scrubber 2.5–3 min CONFIRMED; overall 30 min ASSUMED. S0 models recovery-bearing transfers with no same-bucket passthrough, cumulative matured-mass availability and a minute dependency	BT→L1→TBT and Low-Cd→L2→BT are ASSUMED; availability, residence validation and shared-tank/filter constraints remain TBC	Operating timestamps, output plan, availability and approved campaign/concurrency rules
130/140LG product storage	383,024 t; solver-backed 130LG-ST-01 and 140LG-RC-01/-02; documented RL1/RL2 → flip gate → RL3/RL4 → outbound BL	Physical product arrangement, 130LG/SL11 rate (1,000 TPH S0 envelope ASSUMED), pairing and confluence TBC. BoD says one product reclaimer; tagged description says two. C1 is a project/model binding, not a PDF label	Arrangement, live SMS/SQMS, stacker rating, reach matrix, confluence capacity and operating grade map
Option boundary	Grey `TO DIGUE (OPTION-2)` exists	Phase-1 selection	Formal option decision; excluded meanwhile

13.6 Control domains

Domain	Coordinated scope	Why separate
A · Phosphate	Louta+C1+Benguerir+C2+M16→PAP/drying/SPH; one plan, three SMS outputs	Shared quality, inventory, corridors and M16 resources
B · Sulphur	M17 closed-store replenishment/reclaim→melting	No shared material, machines or blend constraints

Both use one configurable engine and UI but solve independently. New areas/SMS are added through configuration.

13.7 Current implementation baseline — September 2026

This subsection records the current application behaviour and supersedes any earlier prototype wording in this scope that implies a single-day horizon, fixed one-hour job duration, a separate contingency-entry table or direct two-stream reclaim blending.

Area	Implemented behaviour
Planning envelope	Default tactical horizon is **7 days / 168 hourly decisions**. Optional rolling refinement uses 15-minute decisions in the first four hours and hourly decisions thereafter. Outage, derate and execution boundaries retain exact minutes, and adjacent compatible slices are merged into one continuous job rather than displayed as artificial one-hour jobs.
Demand	One dated seven-day table covers Benguerir, Louta and MPH. A unit may have **multiple grade/quantity campaigns on the same day**; each row carries an intraday start/end and priority. Demand is retained when a route is unavailable and is reported as unmet rather than silently removed.
Routing and contingencies	Plan → Routing availability reuses the Monitor's global flow diagram with All/Louta/Benguerir/MPH filters. Clicking a planner-controlled machine, conveyor, route or plant opens one harmonised control: **Available / Disabled / Future**, plus a physical-rate cap, applied for the whole run or an exact timed interval. This consolidates planned contingencies; a separate table-entry page is no longer required. Current state is explicit on the icon/line and in the consolidated control list.
Configuration	The simplified Configure section includes one Grade configuration page with Chemical grades, Yield & moisture, Material profiles and Receiving limits. Grade records can be added or deleted persistently; a referenced grade must first be remapped, preventing orphaned material profiles or stockpile assignments. The user-facing nominal grade name is **BT**. PAP-2 exposes the same three lifecycle states as every destination and defaults to **Future**.
Mass and quality basis	Capacities, inventories, rates, jobs and blend tonnes are physical/as-handled values at the applicable stage. Raw physical mass → TSM uses `(1 − raw moisture) × beneficiation yield`; wet washed product retains configured moisture; after drying, 1 physical dry tonne = 1 TSM. Chemistry of a blend is dry-solids weighted, while moisture is physical-mass weighted. TSM appears as a secondary label/hover value only when a valid grade conversion exists.
Integrated mimic and inspection	The same diagram is used for planning and monitoring. Reduced labels preserve legibility while machines, plants and conveyors carry prominent state cues. Stockpile fill is drawn proportional to physical occupancy; hover exposes stockpile grade composition, physical mass, capacity/fill, stage and applicable TSM equivalent.
Job schedule and results	Monitor exposes readiness, demand/service and unmet quantity, quality/off-spec/give-away, inventory/safety/capacity, route and machine utilisation/coverage, positioning/conflict, and replanning/recovery evidence. It also shows the active configurable objective weights. Schedule filters include site/domain, machine, operation and status plus **Displayed period = full run or an exact calendar day**.
M16 blending	The implemented point is the **M16 humid stockpile HP-07** before its PAP receiving campaign. The dedicated Blending page offers **Blend matrix** (operator-set fixed shares) and **Optimized shares** (component masses are solver variables). In optimized mode composition, stacking time, routes, equipment calendars, source inventory, competing work and destination interval are solved together in physical tonnes. The proposal may change on replan until stacking starts; executed stacking locks the resulting matrix. Reclaim cannot start before the configured pile-ready boundary, and stacked batch mass must equal matrix mass exactly.

The following blending extensions are **not yet implemented**: Benguerir washed-product piles into a train campaign; multi-quality raw-stock feed into washing lines; and spatial tracking of layers/zones and their changing reclaim composition. Grade-class chemistry is currently configured as vectors plus destination receiving intervals; defining the grades themselves entirely by min/max characteristic intervals remains a planned modelling extension.

> **Rev AC verification history:** the focused blending/scheduling regression set passes 61 tests. A full seven-day physical solve has also returned a feasible 12,000 t HP-07 campaign with jointly selected 1,000 t TBT + 11,000 t BT-LOW-CD and no reclaim before the 48-hour ready boundary. These are deterministic simulator checks, not proof of plant performance. The wider legacy repository suite still contains stale UI/report expectations and is not yet a clean acceptance suite; operational validation continues to require governed SMS/SQMS data, approved quality intervals, route evidence and shadow/pilot comparison.

13.8 Review hardening and business-case comparison — Rev AD

The report-aligned application was recovered from the Desktop OneDrive backup into `Claude-Integrated-App/simulator`. The main launcher now targets this copy. The earlier simulator, original backup, saved grade catalogue and audit history remain preserved; recovery provenance is recorded in `generated_outputs/project_review/recovery_manifest.json`.

Area	Implemented review improvement
Blending and execution	Fixed chemistry follows actual integer-tonne allocation; missing assays stay unknown through stock mixing, snapshots and in-transit parcels. First actual stacking locks the matrix, and reclaim requires physical component completion, readiness and hard quality. Invalid timing and unavailable upstream production withhold dependent work.
Demand and grades	Atomic dated edits preserve hourly profiles and C2 allocation by default. Real clock offsets, raw/washed compatibility, referenced-grade deletion and finite catalogue values are checked. Per-campaign priority is retained; blank inherits the receiving-unit value. Configure again exposes inventory snapshots, and fresh scenarios preserve plans during navigation.
KPIs and weights	Terminal service is credited inside due buckets; process feed inside daily grade/priority campaigns. Exact profiles and selected blend chemistry drive quality, with on-spec mass, unknown tonnes, terminal versus handled mass and physical safety exposure separated. Balanced, Quality and Inventory continuity weight profiles are optional, uncalibrated sensitivities.
Independent SMS comparison	A synthetic 12-hour three-site case uses three deterministic local protocols, equal nominations and the same physical replay as the coordinated plan. A finite 500 t assumed receiving buffer retains delayed arrivals. The strongest local rule is the comparator; errors, negative results and null controls are retained.
Observed demonstration result	With the assumed receiving interruption, both the strongest local rule and the model deliver 600 t on-time/on-spec. Coordinated staging falls from 200 to 0 t peak and from 750 to 0 t·h holding. There is **zero demonstrated throughput uplift** in this case; neither staging nor annual savings are claimed for the actual plant.
Full-project sensitivity	Copied current, demand-peak, outage, quality-stress and low-stock cases compare local source portfolios with coordinated allocation. These arms share the global feasibility scheduler and therefore do not isolate full-plant information blindness. Failed solves or execution checks cannot support monetary benefit.

Detailed findings, reproducible experiments, verification logs and the economic evidence plan are delivered in `generated_outputs/project_review/MPH-Project-Review-and-Business-Case.pdf`. The separate review records current test results; prior Rev AC counts below are historical. A numerical plant ROI still requires approved snapshots, operator protocols, holding/backpressure behavior, matched replay and governed costs.

Abbreviations

Abbrev.	Meaning	Abbrev.	Meaning
ACS	Anti-Collision System	ASP	Advanced Stockyard Planning
BPL	Bone Phosphate of Lime	CMS	Maintenance / Asset system
DCS	Distributed Control System	GA	General Arrangement
HMI	Human-Machine Interface	ICD	Interface Control Document
LIMS	Laboratory Information Management System	MES	Manufacturing Execution System
MPH	Mzinda Phosphate Hub	MTPY	Million tonnes per year
OPC UA/HDA	OPC Unified Architecture / Historical Data Access	PAP	Phosphoric Acid Plant
PFD	Process Flow Diagram	PLC	Programmable Logic Controller
PSD	Particle-Size Distribution	SMS/SQMS	Stockpile (Quality) Management System
SPH	Safi Platform Hub	TBE	Technical Bid Evaluation
TBC/TBD	To Be Confirmed / Determined	TPH	Tonnes per hour
TSM	Zero-moisture merchant dry mass		